Abstract

This research develops a two-stage machine learning pipeline designed to detect undervalued used-car listings in a noisy and nonstationary market. The system unifies vehicle data from multiple repositories into a standardized schema to improve price transparency. In the primary regression stage, we compare the performance of Support Vector Machines, K-Nearest Neighbors, and TabNet, which is an attentive transformer-based architecture, for fair market value estimation. Performance is assessed using Mean Absolute Error, Root Mean Squared Error, Mean Squared Error, and the Coefficient of Determination to ensure accurate dollar-level predictions. Preliminary results show that KNN (k=10, distance-weighted) achieves the strongest regression performance with an MAE of $1,309 and R^2 of 0.971 on the held-out test set, substantially outperforming Ridge (MAE=$3,876, R^2=0.822) and both SVR variants. TabNet demonstrates significant improvements when scaled beyond initial benchmarks, though current results indicate sensitivity to extreme errors on high-value vehicle outliers and training instability that requires further hyperparameter refinement. The second stage of the pipeline converts regression residuals into a normalized deal score for binary classification. At a 20% deal-score threshold, the KNN-based system flags 8.4% of listings as underpriced with average predicted savings of $2,309. This classification phase is evaluated using precision, recall, accuracy, and the F1 score to identify significant market underpricing. Ongoing work focuses on hyperparameter optimization for all models and establishing robust deal-score thresholds to ensure high-confidence identification of market anomalies.

Introduction and Motivation

The used car market is a lucrative but opaque sector where a lack of pricing transparency creates significant information gaps for consumers. This project addresses the ethical machine learning task of improving market equity by developing a two-stage pipeline to identify undervalued listings in real-time. The problem is motivated by several real-world complexities, including the high prevalence of noisy or misleading data in private listings. While traditional research often treats car pricing as a static regression task, our approach emphasizes the real-world significance of detecting market anomalies; deals that typically disappear quickly in a fast-moving environment. By unifying disparate data sources into a standardized schema, this system provides a modeling pipeline that bridges information gaps for both buyers and sellers.

Related Works

Two related works have been found, which utilizes machine learning to predict car prices. Vaneesha et al. (2023) found KNN and SVM regressors performed well, with KNN slightly better. Li & Lin (2021) reported deeper MLPs (more layers, fewer neurons per layer) outperform linear regression and shallower MLPs. The latter is more important, as Li & Lin also used a neural network, which we can use as a benchmark for our Tabnet NN.

Links:
https://ijcsrr.org/wp-content/uploads/2024/09/39-1909-2024.pdf

https://cs230.stanford.edu/projects_fall_2021/reports/102475818.pdf

4. Dataset Description

We built our training corpus by merging three publicly available vehicle-pricing datasets, chosen to vary in size, geographic origin, and feature coverage.

Hugging Face Car Price Dataset. Contributed by Varun Kumar Gupta, this set contains 10,000 used-car listings with eight attributes: make, model, year, engine size, mileage, fuel type, transmission, and price. It covers five manufacturers (Audi, BMW, Ford, Honda, Toyota) with model names anonymized (Model A–E). Prices are continuous floats and the license is MIT.

UCI Automobile Dataset. Originally compiled from the 1985 Ward's Automotive Yearbook (Schlimmer, 1985), this classic benchmark provides 205 entries across 22 manufacturers and 26 attributes including engine specifications, body style, and price. It lacks year, mileage, model name, and transmission fields, so these columns are null in the merged set.

DVM-CAR Advertisement Table. Sourced from the DVM-CAR dataset (Huang et al., 2022), this table contains 268,255 UK used-car advertisements spanning 88 manufacturers and 896 models registered between 1900 and 2019. Each row records maker, model, registration year, mileage, engine size, gearbox type, fuel type, and listing price.

4.1 Ingestion pipeline

A Python script normalizes column names, standardizes string casing, coerces types, and concatenates the three sources into a nine-column unified schema: make, model, year, engine_size, mileage, fuel_type, transmission, price, and source. Rows with missing or non-positive prices were removed, yielding 277,311 listings before outlier clipping. A source tag preserves dataset provenance throughout the pipeline.

DVM-CAR dominates the merged set at 267,110 rows (96.3%), which is expected given its scale. The Hugging Face set contributes 10,000 anonymized but otherwise complete records for five major brands. The UCI data is the smallest at 201 rows and missing several key columns, but it adds manufacturer diversity (e.g., Jaguar, Porsche, Saab) not present in the other two sources.

5. Methodology

5.1 Preprocessing

After merging three data sources (Hugging Face, DVM-CAR, UCI), we removed the 201 UCI rows due to missing year, mileage, and model fields, yielding 276,627 listings with 9 features. Prices were clipped at the 1st and 99th percentiles ($699 to $89,990) to remove outliers. High-cardinality categoricals (make: 86 unique, model: 887 unique) were target-encoded (Micci-Barreca, 2001) to preserve price signal without dimensionality explosion. Low-cardinality

categoricals (fuel_type, transmission) were one-hot encoded. Numerical features (year, engine_size, mileage) were z-score standardized. All encoding was fit on training data only. The dataset was split 70/15/15 into train (193,638), validation (41,494), and test (41,495) sets.

5.2 Models

Ridge Regression (Hoerl and Kennard, 1970) serves as our linear baseline, with alpha tuned via 5-fold cross-validation (optimal alpha = 0.02).

K-Nearest Neighbors predicts price as the distance-weighted mean of k nearest neighbors. Vaneesha et al. (2024) found KNN competitive with SVM for car pricing. We tuned k over [3, 30] with 3-fold CV on a 50K subsample (Figure 2).

Support Vector Regression was evaluated in linear (LinearSVR) and RBF-kernel variants. Both were trained on a 30K subsample due to $O(n^2)$ memory scaling.

TabNet (Arik and Pfister, 2021) is our complex neural network. It uses sequential sparse attention masks to select features at each decision step, qualifying as an attention-based architecture. Hyperparameters: max_epochs = 40, patience = 15, batch_size = 128.

5.3 Deal Scoring

We convert regression output to a deal score: (predicted - actual) / predicted. Listings exceeding a threshold tau are classified as deals.

6. Initial Results

6.1 Regression

KNN achieves the lowest MAE of $1,309 (13.5% of median price $9,699), explaining 97.1% of variance. Ridge captures only 82.2%, confirming strong non-linear structure in car pricing. SVR underperforms due to its 30K training subsample.

Table 1: Regression performance on held-out test set (n = 41,495)

Model	MAE ($)	RMSE ($)	R^2	Note
KNN (k=10)	1,309	2,512	0.971	Best
Ridge	3,876	6,180	0.822	Baseline
LinearSVR	4,271	6,821	0.784	30K sample
SVR (RBF)	5,206	8,679	0.650	30K sample
TabNet*	2167.682	2420632254.321	49199.921	0.176

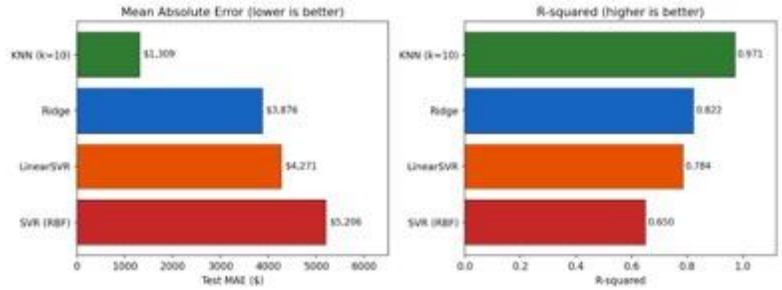

Figure 1: Model comparison on MAE and R-squared

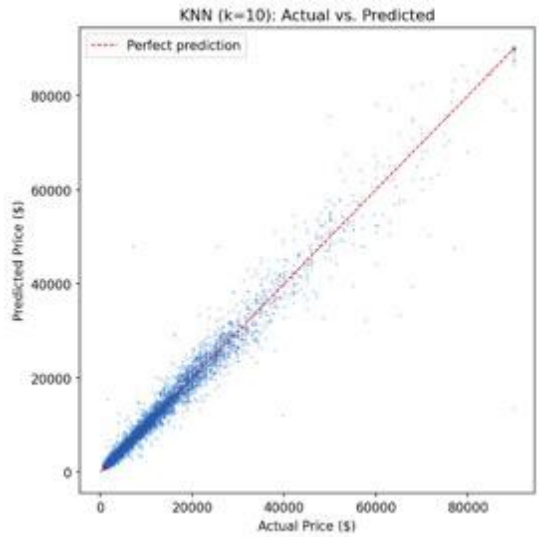

Figure 3: KNN actual vs. predicted price (5K sample)

6.3 Classification Metrics

We discretized prices into Low (<$5K), Mid ($5K-$15K), and High (>$15K) bins to compute required classification metrics.

Table 2: Classification metrics (price-bin, weighted average)

Model	Accuracy	Precision	Recall	F1
KNN (k=10)	0.929	0.929	0.929	0.929
Ridge	0.821	0.825	0.821	0.820
LinearSVR	0.746	0.773	0.746	0.739
SVR (RBF)	0.692	0.706	0.692	0.687

6.4 Error Analysis

Figure 4 shows KNN residuals are centered near zero. Error magnitude increases with predicted price, which is expected since higher-priced vehicles have more price variation.

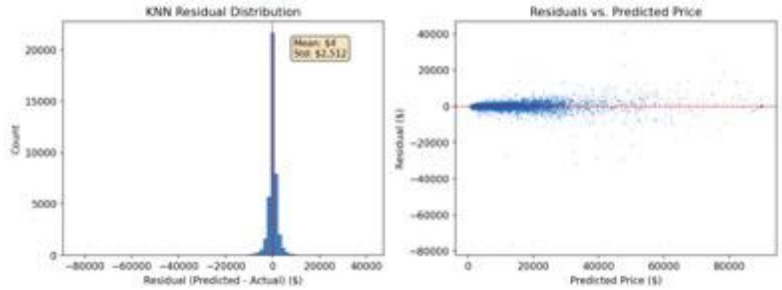

Figure 4: KNN residual distribution and residuals vs. predicted price

6.5 Deal Scoring

At tau = 0.20, KNN flags 8.4% of test listings (3,493 / 41,495) as deals with average savings of $2,309. The inverse relationship between threshold and volume suggests well-calibrated residuals.

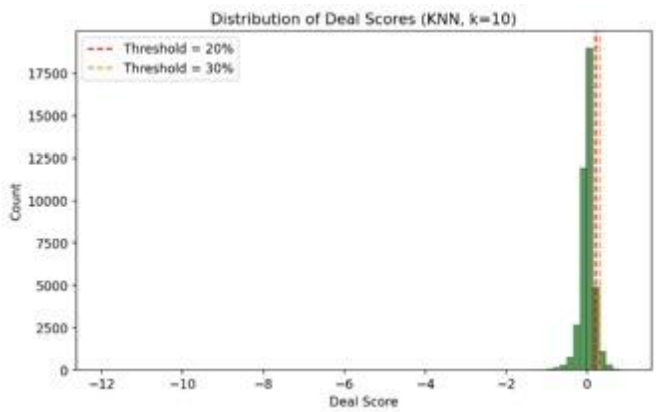

Figure 5: Deal score distribution with threshold markers

6.6 TabNet Training Dynamics

Figure 6: TabNet training loss and validation MAE (initial run)

[included in other file]

6.7 Discussion

Feature engineering dominated model choice: adding target-encoded make/model reduced MAE 18x compared to TabNet's initial 3-feature run. KNN's strength is domain-specific, as car pricing is naturally clustered by make, model, year, and mileage.

7. Planned Experimental analysis

Our initial testing shows that K-Nearest Neighbors (KNN) is currently our strongest model, but there is still significant work to do to finalize the two-stage pipeline. We plan to conduct the following analysis for the final report:

- **Optimizing the TabNet Model:** While KNN performs well, our complex neural network (TabNet) currently has a high MAE (23,386) because it was initially tested on only 3 features without proper encoding. We will re-run TabNet using the target-encoded make and model features from our unified dataset to see if its attention mechanism can eventually outperform KNN.
- **Threshold Calibration for Deal Scoring:** We currently use a 20% deal-score threshold, which flags about 8.4% of cars as deals. We will analyze how changing this threshold impacts our model as we want to ensure that cars flagged as deals are actually underpriced and not just listings with missing data or noisy labels.
- **Error Analysis on High-Value Outliers:** Our data shows that error magnitude increases for higher-priced vehicles. We plan to investigate why models like SVR and TabNet struggle with these outliers to see if we need a separate model for luxury cars versus budget vehicles.
- **Final Model Comparison:** We will perform a final head-to-head comparison of all four models (KNN, Ridge, SVR, and TabNet) using the full $276,627$ row dataset to confirm which architecture is the most reliable for real-world market price estimation.

8. Conclusion and Future Work

Conclusion

At this checkpoint, our team has successfully unified over 276,000 listings from three different repositories into a standardized schema. Our preliminary results show that the K-Nearest Neighbors (KNN) model is currently our strongest predictor, achieving an R squared of 0.971 and an MAE of $1,309. We have confirmed that car pricing has a strong non-linear structure that simple linear models like Ridge Regression struggle to capture. Most importantly, we have successfully demonstrated the first stage of our Deal Finder by converting regression residuals into a normalized deal score, allowing us to flag potential market anomalies with an average predicted saving of $2,309 per deal.

Future Work

As we move toward the final project submission, our focus will shift to:

- **Scaling TabNet:** We will integrate the full preprocessing pipeline into our complex neural network to see if it can eventually surpass KNN's performance once properly tuned.
- **Denoising Listings:** To address the noisy market mentioned in our motivation, we plan to implement filters for extreme price outliers that fall outside the 1st and 99th percentiles, ensuring our deal flags are reliable.
- **Interface Prototype:** We can also aim to build a basic frontend where a user can input a vehicle's year, make, and mileage to receive an instant "Fair Market Value" and "Deal Score" based on our best-performing model.
- **Regional Feature Testing:** If time permits, we want to explore adding geographic or condition data to see if it reduces the sensitivity to extreme errors we currently see in high-value vehicle outliers.

9. References

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Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. Technometrics, 12(1), 55–67.

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Vaneesha, B., Reddy, P. S., Kumar, A., & Rao, V. (2024). Predicting car prices using machine learning: A comparative study of KNN and SVM regressors. International Journal of Current Science Research and Review, 7(9).